# RAJALAKSHMI ENGINEERING COLLEGE

**RAJALAKSHMI NAGAR, THANDALAM – 602 105**




# CS23331 - DESIGN AND ANALYSIS OF ALGORITHMS

**Laboratory Record Note Book**

| | |
|---|---|
| Name : | BENNO JEFRINA J. ..... |
| Year / Branch : | 2ND YEAR, INFORMATION TECHNOLOGY, SEC -A ..... |
| University Register No. : | 2116251001028 ..... |
| College Roll : | 251001028 ..... |
| Semester : | III SEM ..... |
| Academic Year : | 2026-2027 ..... |

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RAJALAKSHMI ENGINEERING COLLEGE

Benno Jefrina J 251001028-2025-IT

B2

| | |
|---|---|
| **Started on** | Thursday, 6 August 2026, 10:13 AM |
| **State** | Finished |
| **Completed on** | Thursday, 6 August 2026, 10:24 AM |
| **Time taken** | 10 mins 41 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
    {
         i++;
         s += i;
     }
}
```

**Note:** No need of counter increment for declarations and scanf() and count variable printf() statements.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**For example:**

| Input | Result |
|---|---|
| 9 | 12 |

**Answer:** (penalty regime: 0 %)

Ace editor not ready. Perhaps reload page?

Falling back to raw text area.

```
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int count=0;
    int i=1;
    count+=1;
    int s=1;
    count+=1;
    while(s<=n){
        count+=1;
        i++;
        count+=1;
        s+=i;
        count+=1;
    }
    count+=1;
    printf("%d",count);
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 9 | 12 | 12 | ✔ |
| ✔ | 4 | 9 | 9 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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B2

| | |
|---|---|
| **Started on** | Thursday, 6 August 2026, 10:28 AM |
| **State** | Finished |
| **Completed on** | Thursday, 6 August 2026, 10:47 AM |
| **Time taken** | 19 mins 1 sec |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
 }
```

**Note:** No need of counter increment for declarations and scanf() and count variable printf() statements.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**Answer:** (penalty regime: 0 %)

```c
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int c=0;
    c++;
    if(n==1){
        c++;
    }
    else{
        c++;
        for(int i=1;i<=n;i++){
            c++;
            c++;
            c++;
            for(int j=1;j<=n;j++){
                c++;
                c++;

                break;

            }
        }
    }

    printf("%d",c);
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 2 | 12 | 12 | ✔ |
| ✔ | 1000 | 5002 | 5002 | ✔ |
| ✔ | 143 | 717 | 717 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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B2

| | |
|---|---|
| **Started on** | Thursday, 6 August 2026, 10:48 AM |
| **State** | Finished |
| **Completed on** | Thursday, 6 August 2026, 10:52 AM |
| **Time taken** | 4 mins 37 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

**Note:** No need of counter increment for declarations and scanf() and counter variable printf() statement.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**Answer:**

```
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int c=0;
    c++;
    int i;
    for(i=1;i<=n;i++){
        c++;
        c++;
        if(n%i==0){
            c++;
        }
    }
    printf("%d",c);
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 12 | 31 | 31 | ✔ |
| ✔ | 25 | 54 | 54 | ✔ |
| ✔ | 4 | 12 | 12 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

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Done

**Opened:** Tuesday, 15 July 2025, 10:08 AM
**Closed:** Saturday, 10 October 2026, 11:16 PM

Attempts allowed: 2

Grading method: Highest grade

## Summary of your previous attempts

| Attempt | State | Marks / 1.00 | Grade / 10.00 | Review |
|---|---|---|---|---|
| 1 | Finished<br>Submitted Thursday, 6 August 2026, 11:00 AM | 1.00 | 10.00 | Review |

## Your final grade for this quiz is 10.00/10.00.

Back to the course

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Data retention summary

RAJALAKSHMI ENGINEERING COLLEGE

Benno Jefrina J 251001028-2025-IT

B2

| | |
|---|---|
| **Started on** | Thursday, 6 August 2026, 11:00 AM |
| **State** | Finished |
| **Completed on** | Thursday, 6 August 2026, 11:04 AM |
| **Time taken** | 3 mins 59 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

**Note:** No need of counter increment for declarations and scanf() and count variable printf() statements.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**Answer:**

```
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int c=0;
    c++;
    int rev=0;
    c++;
    int remainder;
    while(n!=0){
        c++;
        remainder=n%10;
        c++;
        rev=rev*10+remainder;
        c++;
        n/=10;
        c++;
    }
    c++;
    printf("%d",c);
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 12 | 11 | 11 | ✔ |
| ✔ | 1234 | 19 | 19 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Benno Jefrina J 251001028-2025-IT

B2

| | |
|---|---|
| **Started on** | Thursday, 27 August 2026, 10:09 AM |
| **State** | Finished |
| **Completed on** | Thursday, 27 August 2026, 10:20 AM |
| **Time taken** | 11 mins 6 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Write a program to take value V and we want to make change for V Rs, and we have infinite supply of each of the denominations in Indian currency, i.e., we have infinite supply of { 1, 2, 5, 10, 20, 50, 100, 500, 1000} valued coins/notes, what is the minimum number of coins and/or notes needed to make the change.

Input Format:

Take an integer from stdin.

Output Format:

print the integer which is change of the number.

Example Input :

64

Output:

4

Explanaton:

We need a 50 Rs note and a 10 Rs note and two 2 rupee coins.

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>
int main() {
    int V;
    int count = 0;
    int notes[] = {1000, 500, 100, 50, 20, 10, 5, 2, 1};
    int n = 9;
    if (scanf("%d", &V) != 1) {
        return 0;
    }
    for (int i = 0; i < n; i++) {
        if (V >= notes[i]) {
            count += V / notes[i];
            V = V % notes[i];
        }
    }
    printf("%d\n", count);
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 49 | 5 | 5 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Done

**Opened:** Wednesday, 15 July 2026, 10:00 AM
**Closed:** Tuesday, 20 October 2026, 11:56 AM

Attempts allowed: 3

Grading method: Highest grade

## Summary of your previous attempts

| Attempt | State | Marks / 1.00 | Grade / 10.00 | Review |
|---|---|---|---|---|
| 1 | Finished<br>Submitted Thursday, 27 August 2026, 10:32 AM | 1.00 | 10.00 | Review |

## Your final grade for this quiz is 10.00/10.00.

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B2

| | |
|---|---|
| **Started on** | Thursday, 27 August 2026, 10:34 AM |
| **State** | Finished |
| **Completed on** | Monday, 31 August 2026, 9:03 AM |
| **Time taken** | 3 days 22 hours |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

A person needs to eat burgers. Each burger contains a count of calorie. After eating the burger, the person needs to run a distance to burn out his calories.

If he has eaten i burgers with c calories each, then he has to run at least $3^i * c$ kilometers to burn out the calories. For example, if he ate 3

burgers with the count of calorie in the order: [1, 3, 2], the kilometers he needs to run are $(3^0 * 1) + (3^1 * 3) + (3^2 * 2) = 1 + 9 + 18 = 28$.

But this is not the minimum, so need to try out other orders of consumption and choose the minimum value. Determine the minimum distance

he needs to run. Note: He can eat burger in any order and use an efficient sorting algorithm.Apply greedy approach to solve the problem.

**Input Format**

First Line contains the number of burgers

Second line contains calories of each burger which is n space-separate integers

**Output Format**

Print: Minimum number of kilometers needed to run to burn out the calories

**Sample Input**

3

5 10 7

**Sample Output**

76

**For example:**

| Test | Input | Result |
|---|---|---|
| Test Case 1 | 3<br>1 3 2 | 18 |

**Answer:** (penalty regime: 0 %)

```c
#include<stdio.h>
#include<math.h>
int main(){
    int n;
    scanf("%d",&n);
    int arr[n];
    for(int i=0;i<n;i++){
        scanf("%d",&arr[i]);
    }
    for (int i = 0; i < n - 1; i++) {
        for (int j = 0; j < n - i - 1; j++) {
            if (arr[j] < arr[j + 1]) {
                int temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
            }
        }
    }
    int s=0;
    for(int i=0;i<n;i++){
        s+=(pow(n,i))*arr[i];
    }
    printf("%d",s);
}
```

| | Test | Input | Expected | Got | |
|---|---|---|---|---|---|
| ✔ | Test Case 1 | 3<br>1 3 2 | 18 | 18 | ✔ |
| ✔ | Test Case 2 | 4<br>7 4 9 6 | 389 | 389 | ✔ |
| ✔ | Test Case 3 | 3<br>5 10 7 | 76 | 76 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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B2

| | |
|---|---|
| **Started on** | Thursday, 27 August 2026, 11:22 AM |
| **State** | Finished |
| **Completed on** | Thursday, 27 August 2026, 11:26 AM |
| **Time taken** | 4 mins 22 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Given an array of N integer, we have to maximize the sum of arr[i] * i, where i is the index of the element (i = 0, 1, 2, ..., N).Write an algorithm based on Greedy technique with a Complexity O(nlogn).

Input Format:

First line specifies the number of elements-n

The next n lines contain the array elements.

Output Format:

Maximum Array Sum to be printed.

Sample Input:

5

2 5 3 4 0

Sample output:

40

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int main() {
    int n;
    if (scanf("%d", &n) != 1 || n <= 0) return 0;

    // Allocate memory for array
    long long *arr = (long long *)malloc(n * sizeof(long long));
    for (int i = 0; i < n; i++) {
        scanf("%lld", &arr[i]);
    }

    // Short Inline Shell Sort (O(n log n) average speed)
    for (int gap = n / 2; gap > 0; gap /= 2) {
        for (int i = gap; i < n; i++) {
            long long temp = arr[i];
            int j;
            for (j = i; j >= gap && arr[j - gap] > temp; j -= gap) {
                arr[j] = arr[j - gap];
            }
            arr[j] = temp;
        }
    }

    // Calculate maximum sum
    long long max_sum = 0;
    for (int i = 0; i < n; i++) {
        max_sum += arr[i] * i;
    }

    // Print output
    printf("%lld\n", max_sum);

    free(arr);
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 5<br>2<br>5<br>3<br>4<br>0 | 40 | 40 | ✔ |
| ✔ | 10<br>2<br>2<br>2<br>4<br>4<br>3<br>3<br>5<br>5<br>5 | 191 | 191 | ✔ |
| ✔ | 2<br>45<br>3 | 45 | 45 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Benno Jefrina J 251001028-2025-IT

B2

| | |
|---|---|
| **Started on** | Thursday, 27 August 2026, 11:27 AM |
| **State** | Finished |
| **Completed on** | Thursday, 27 August 2026, 11:31 AM |
| **Time taken** | 3 mins 47 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Given two arrays array_One[] and array_Two[] of same size N. We need to first rearrange the arrays such that the sum of the product of pairs( 1 element from each) is minimum. That is SUM (A[i] * B[i]) for all i is minimum.

**For example:**

| Input | Result |
|---|---|
| 3<br>1<br>2<br>3<br>4<br>5<br>6 | 28 |

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

    int arrayOne[n];
    for (int i = 0; i < n; i++) {
        if (scanf("%d", &arrayOne[i]) != 1) return 0;
    }

    int arrayTwo[n];
    for (int i = 0; i < n; i++) {
        if (scanf("%d", &arrayTwo[i]) != 1) return 0;
    }

    int i, j, temp;
    long long minProductSum = 0;

    for (i = 0; i < n - 1; i++) {
        for (j = 0; j < n - i - 1; j++) {
            if (arrayOne[j] > arrayOne[j + 1]) {
                temp = arrayOne[j];
                arrayOne[j] = arrayOne[j + 1];
                arrayOne[j + 1] = temp;
            }
        }
    }

    for (i = 0; i < n - 1; i++) {
        for (j = 0; j < n - i - 1; j++) {
            if (arrayTwo[j] < arrayTwo[j + 1]) {
                temp = arrayTwo[j];
                arrayTwo[j] = arrayTwo[j + 1];
                arrayTwo[j + 1] = temp;
            }
        }
    }

    for (i = 0; i < n; i++) {
        minProductSum += (long long)arrayOne[i] * arrayTwo[i];
    }

    printf("%lld\n", minProductSum);

    return 0;
}

```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 3<br>1<br>2<br>3<br>4<br>5<br>6 | 28 | 28 | ✔ |
| ✔ | 4<br>7<br>5<br>1<br>2<br>1<br>3<br>4<br>1 | 22 | 22 | ✔ |
| ✔ | 5<br>20<br>10<br>30<br>10<br>40<br>8<br>9<br>4<br>3<br>10 | 590 | 590 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Benno Jefrina J 251001028-2025-IT

B2

| | |
|---|---|
| **Started on** | Thursday, 20 August 2026, 10:45 AM |
| **State** | Finished |
| **Completed on** | Thursday, 20 August 2026, 10:48 AM |
| **Time taken** | 3 mins 6 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

**Problem Statement**

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

**Answer:** (penalty regime: 0 %)

```
#include<stdio.h>
int main(){
int n;
scanf("%d",&n);
int a[n];
for(int i=0;i<n;i++){
scanf("%d",&a[i]);
}
int c=0;
for(int i=0;i<n;i++){
if(a[i]==0){
c++;}
}
printf("%d",c);
return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 5<br>1<br>1<br>1<br>0<br>0 | 2 | 2 | ✔ |
| ✔ | 10<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1 | 0 | 0 | ✔ |

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 8<br>0<br>0<br>0<br>0<br>0<br>0<br>0<br>0 | 8 | 8 | ✔ |
| ✔ | 17<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>0<br>0 | 2 | 2 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

Benno Jefrina J 251001028-2025-IT

B2

| | |
|---|---|
| **Started on** | Thursday, 20 August 2026, 10:49 AM |
| **State** | Finished |
| **Completed on** | Thursday, 20 August 2026, 11:02 AM |
| **Time taken** | 13 mins 32 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than `⌊n / 2⌋` times. You may assume that the majority element always exists in the array.

**Example 1:**

```
Input: nums = [3,2,3]
Output: 3
```

**Example 2:**

```
Input: nums = [2,2,1,1,1,2,2]
Output: 2
```

**Constraints:**

- `n == nums.length`
- `1 <= n <= 5 * 10^4`
- `-2^31 <= nums[i] <= 2^31 - 1`

**For example:**

| Input | Result |
|---|---|
| 3<br>3 2 3 | 3 |
| 7<br>2 2 1 1 1 2 2 | 2 |

**Answer:** (penalty regime: 0 %)

```c
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int a[n];
    for(int i=0;i<n;i++){
        scanf("%d",&a[i]);
    }
    int temp=0;
    for(int i=0;i<n;i++){
        if(a[i]>temp){
            temp=a[i];
        }
    }
    printf("%d",temp);
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 3<br>3 2 3 | 3 | 3 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

Benno Jefrina J 251001028-2025-IT

B2

| | |
|---|---|
| **Started on** | Thursday, 20 August 2026, 11:11 AM |
| **State** | Finished |
| **Completed on** | Thursday, 20 August 2026, 11:36 AM |
| **Time taken** | 25 mins 3 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

**Problem Statement:**

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

**Input Format**

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

**Output Format**

First Line Contains Integer – Floor value for x

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>
int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;
    int a[n];
    for (int i = 0; i < n; i++) {
        if (scanf("%d", &a[i]) != 1) return 0;
    }

    int x;
    if (scanf("%d", &x) != 1) return 0;

    int low = 0;
    int high = n - 1;
    int floor_val = -1;

    while (low <= high) {
        int mid = low + (high - low) / 2;

        if (a[mid] == x) {
            floor_val = a[mid];
            break;
        }
        else if (a[mid] < x) {
            floor_val = a[mid];
            low = mid + 1;
        }
        else {
            high = mid - 1;
        }
    }

    printf("%d\n", floor_val);

    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 6<br>1<br>2<br>8<br>10<br>12<br>19<br>5 | 2 | 2 | ✔ |

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 5<br>10<br>22<br>85<br>108<br>129<br>100 | 85 | 85 | ✔ |
| ✔ | 7<br>3<br>5<br>7<br>9<br>11<br>13<br>15<br>10 | 9 | 9 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Benno Jefrina J 251001028-2025-IT

B2

| | |
|---|---|
| **Started on** | Thursday, 27 August 2026, 11:04 AM |
| **State** | Finished |
| **Completed on** | Thursday, 27 August 2026, 11:13 AM |
| **Time taken** | 8 mins 43 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

**Problem Statement:**

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

**Input Format**

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

**Output Format**

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n, x;
    scanf("%d", &n);

    int a[n];

    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);

    scanf("%d", &x);

    int left = 0, right = n - 1;
    int found = 0;

    while (left < right) {
        int sum = a[left] + a[right];

        if (sum == x) {
            printf("%d\n", a[left]);
            printf("%d\n", a[right]);
            found = 1;
            break;
        } else if (sum < x) {
            left++;
        } else {
            right--;
        }
    }

    if (!found)
        printf("No\n");

    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 4<br>2<br>4<br>8<br>10<br>14 | 4<br>10 | 4<br>10 | ✔ |

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 5<br>2<br>4<br>6<br>8<br>10<br>100 | No | No | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

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B2

| | |
|---|---|
| **Started on** | Thursday, 27 August 2026, 11:13 AM |
| **State** | Finished |
| **Completed on** | Thursday, 27 August 2026, 11:21 AM |
| **Time taken** | 7 mins 55 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:
The first line contains the no of elements in the list-n
The next n lines contain the elements.

Output:
Sorted list of elements

**For example:**

| Input | Result |
|---|---|
| 5<br>67 34 12 98 78 | 12 34 67 78 98 |

**Answer:**

```c
#include <stdio.h>

int main() {
    int n, temp;

    if (scanf("%d", &n) != 1) return 0;
    int arr[n];

    for (int i = 0; i < n; i++) {
        if (scanf("%d", &arr[i]) != 1) return 0;
    }

    for (int i = 0; i < n - 1; i++) {
        for (int j = 0; j < n - i - 1; j++) {
            if (arr[j] > arr[j + 1]) {
                temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
            }
        }
    }

    for (int i = 0; i < n; i++) {
        printf("%d ", arr[i]);
    }
    printf("\n");

    return 0;
}

```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 5<br>67 34 12 98 78 | 12 34 67 78 98 | 12 34 67 78 98 | ✔ |
| ✔ | 10<br>1 56 78 90 32 56 11 10 90 114 | 1 10 11 32 56 56 78 90 90 114 | 1 10 11 32 56 56 78 90 90 114 | ✔ |
| ✔ | 12<br>9 8 7 6 5 4 3 2 1 10 11 90 | 1 2 3 4 5 6 7 8 9 10 11 90 | 1 2 3 4 5 6 7 8 9 10 11 90 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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B2

| | |
|---|---|
| **Started on** | Thursday, 17 September 2026, 11:14 AM |
| **State** | Finished |
| **Completed on** | Thursday, 17 September 2026, 11:17 AM |
| **Time taken** | 3 mins 16 secs |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 10.00 out of 10.00

**Playing with Numbers:**

Ram and Sita are playing with numbers by giving puzzles to each other. Now it was Ram term, so he gave Sita a positive integer 'n' and two numbers 1 and 3. He asked her to find the possible ways by which the number n can be represented using 1 and 3.Write any efficient algorithm to find the possible ways.

**Example 1:**

***Input:*** *6*
***Output:****6*
***Explanation:*** *There are 6 ways to 6 represent number with 1 and 3*

*1+1+1+1+1+1*
*3+3*
*1+1+1+3*
*1+1+3+1*
*1+3+1+1*
*3+1+1+1*

**Input Format**

First Line contains the number n

**Output Format**

**Print: The number of possible ways 'n' can be represented using 1 and 3**

Sample Input

6

Sample Output

6

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n;
    if (scanf("%d", &n) == 1) {
        if (n == 0 || n == 1 || n == 2) {
            printf("1\n");
        } else {
            unsigned long long prev3 = 1;
            unsigned long long prev2 = 1;
            unsigned long long prev1 = 1;
            unsigned long long current = 0;

            for (int i = 3; i <= n; i++) {
                current = prev1 + prev3;
                prev3 = prev2;
                prev2 = prev1;
                prev1 = current;
            }
            printf("%llu\n", current);
        }
    }
    return 0;
}

```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 6 | 6 | 6 | ✔ |
| ✔ | 25 | 8641 | 8641 | ✔ |
| ✔ | 100 | 24382819596721629 | 24382819596721629 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

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B2

| | |
|---|---|
| **Started on** | Thursday, 17 September 2026, 11:20 AM |
| **State** | Finished |
| **Completed on** | Thursday, 17 September 2026, 11:22 AM |
| **Time taken** | 2 mins 19 secs |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 10.00 out of 10.00

**Playing with Chessboard:**

Ram is given with an n*n chessboard with each cell with a monetary value. Ram stands at the (0,0), that the position of the top left white rook. He is been given a task to reach the bottom right black rook position (n-1, n-1) constrained that he needs to reach the position by traveling the maximum monetary path under the condition that he can only travel one step right or one step down the board. Help ram to achieve it by providing an efficient DP algorithm.

**Example:**

**Input**

3

**1** 2 4

**2** 3 4

**8 7 1**

**Output:**

19

**Explanation:**

Totally there will be 6 paths among that the optimal is

Optimal path value:1+2+8+7+1=19

**Input Format**

First Line contains the integer n

The next n lines contain the n*n chessboard values

**Output Format**

Print Maximum monetary value of the path

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

    long long *dp = (long long *)calloc(n, sizeof(long long));
    int *current_row = (int *)malloc(n * sizeof(int));

    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            if (scanf("%d", &current_row[j]) != 1) return 0;
        }

        for (int j = 0; j < n; j++) {
            if (i == 0 && j == 0) {
                dp[j] = current_row[j];
            } else if (i == 0) {
                dp[j] = dp[j-1] + current_row[j];
            } else if (j == 0) {
                dp[j] = dp[j] + current_row[j];
            } else {
                long long top = dp[j];
                long long left = dp[j-1];
                dp[j] = current_row[j] + ((top > left) ? top : left);
            }
        }
    }

    printf("%lld\n", dp[n-1]);

    free(dp);
    free(current_row);

    return 0;
```

```
}

```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 3<br>1 2 4<br>2 3 4<br>8 7 1 | 19 | 19 | ✔ |
| ✔ | 3<br>1 3 1<br>1 5 1<br>4 2 1 | 12 | 12 | ✔ |
| ✔ | 4<br>1 1 3 4<br>1 5 7 8<br>2 3 4 6<br>1 6 9 0 | 28 | 28 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

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B2

| | |
|---|---|
| **Started on** | Thursday, 17 September 2026, 11:22 AM |
| **State** | Finished |
| **Completed on** | Thursday, 17 September 2026, 11:23 AM |
| **Time taken** | 1 min 3 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

| | | | | | | |
|---|---|---|---|---|---|---|
| s1 | a | g | **g** | **t** | **a** | **b** |
| s2 | **g** | x | **t** | **x** | **a** | **y** | **b** |

**The length is 4**

Solveing it using Dynamic Programming

**For example:**

| Input | Result |
|---|---|
| aab<br>azb | 2 |

**Answer:** (penalty regime: 0 %)

```
#include <stdio.h>
#include <string.h>

int main() {
    char s1[1000], s2[1000];
    if (scanf("%s %s", s1, s2) != 2) {
        return 0;
    }

    int m = strlen(s1);
    int n = strlen(s2);
    int dp[m + 1][n + 1];

    for (int i = 0; i <= m; i++) {
        for (int j = 0; j <= n; j++) {
            if (i == 0 || j == 0) {
                dp[i][j] = 0;
            } else if (s1[i - 1] == s2[j - 1]) {
                dp[i][j] = dp[i - 1][j - 1] + 1;
            } else {
                dp[i][j] = (dp[i - 1][j] > dp[i][j - 1]) ? dp[i - 1][j] : dp[i][j - 1];
            }
        }
    }

    printf("%d\n", dp[m][n]);
    return 0;
}

```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | aab<br>azb | 2 | 2 | ✔ |
| ✔ | ABCD<br>ABCD | 4 | 4 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Benno Jefrina J 251001028-2025-IT B2

| | |
|---|---|
| **Started on** | Thursday, 17 September 2026, 11:24 AM |
| **State** | Finished |
| **Completed on** | Thursday, 17 September 2026, 11:24 AM |
| **Time taken** | 52 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n = 9;
    int seq[] = {-1, 3, 4, 5, 2, 2, 2, 2, 3};

    int tails[n];
    int len = 0;

    for (int i = 0; i < n; i++) {
        int low = 0, high = len;
        while (low < high) {
            int mid = low + (high - low) / 2;
            if (tails[mid] <= seq[i]) {
                low = mid + 1;
            } else {
                high = mid;
            }
        }
        tails[low] = seq[i];
        if (low == len) {
            len++;
        }
    }

    printf("%d\n", len);

    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 9<br>-1 3 4 5 2 2 2 2 3 | 6 | 6 | ✔ |
| ✔ | 7<br>1 2 2 4 5 7 6 | 6 | 6 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Benno Jefrina J 251001028-2025-IT

B2

| | |
|---|---|
| **Started on** | Thursday, 17 September 2026, 10:16 AM |
| **State** | Finished |
| **Completed on** | Thursday, 17 September 2026, 10:34 AM |
| **Time taken** | 17 mins 56 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

**For example:**

| Input | Result |
|---|---|
| 5<br>1 1 2 3 4 | 1 |

**Answer:** (penalty regime: 0 %)

```
#include<stdio.h>
int main(){
int n;
scanf("%d",&n);
int a[n];
for(int i=0;i<n;i++){
scanf("%d",&a[i]);
}
for(int i=0;i<n-1;i++){
for(int j=i+1;j<n;j++){
if(a[i]==a[j]){
printf("%d",a[i]);
}
}
}
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 11<br>10 9 7 6 5 1 2 3 8 4 7 | 7 | 7 | ✔ |
| ✔ | 5<br>1 2 3 4 4 | 4 | 4 | ✔ |
| ✔ | 5<br>1 1 2 3 4 | 1 | 1 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Benno Jefrina J 251001028-2025-IT

B2

| | |
|---|---|
| **Started on** | Thursday, 17 September 2026, 10:34 AM |
| **State** | Finished |
| **Completed on** | Thursday, 17 September 2026, 10:36 AM |
| **Time taken** | 2 mins 22 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

**For example:**

| Input | Result |
|---|---|
| 5<br>1 1 2 3 4 | 1 |

**Answer:** (penalty regime: 0 %)

```
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int a[n];
    for(int i=0;i<n;i++){
        scanf("%d",&a[i]);
    }
    for(int i=0;i<n-1;i++){
        for(int j=i+1;j<n;j++){
            if(a[i]==a[j]){
                printf("%d",a[i]);
            }
        }
    }
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 11<br>10 9 7 6 5 1 2 3 8 4 7 | 7 | 7 | ✔ |
| ✔ | 5<br>1 2 3 4 4 | 4 | 4 | ✔ |
| ✔ | 5<br>1 1 2 3 4 | 1 | 1 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Benno Jefrina J 251001028-2025-IT

B2

| | |
|---|---|
| **Started on** | Thursday, 17 September 2026, 10:37 AM |
| **State** | Finished |
| **Completed on** | Thursday, 17 September 2026, 10:44 AM |
| **Time taken** | 7 mins 6 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

**For example:**

| Input | Result |
|---|---|
| 1<br>3 10 17 57<br>6<br>2 7 10 15 57 246 | 10 57 |

**Answer:** (penalty regime: 0 %)

```
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int a;
    scanf("%d",&a);
    int A[a];
    for(int i=0;i<a;i++){
        scanf("%d",&A[i]);
    }
    int b;
    scanf("%d",&b);
    int B[b];
    for(int i=0;i<b;i++){
        scanf("%d",&B[i]);
    }
    for(int i=0;i<a;i++){
        for(int j=0;j<b;j++){
            if(A[i]==B[j]){
                printf("%d ",A[i]);
            }
        }
```

```
    }
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 1<br>3 10 17 57<br>6<br>2 7 10 15 57 246 | 10 57 | 10 57 | ✔ |
| ✔ | 1<br>6 1 2 3 4 5 6<br>2<br>1 6 | 1 6 | 1 6 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Benno Jefrina J 251001028-2025-IT

B2

| | |
|---|---|
| **Started on** | Thursday, 17 September 2026, 10:44 AM |
| **State** | Finished |
| **Completed on** | Thursday, 17 September 2026, 10:49 AM |
| **Time taken** | 4 mins 41 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

**For example:**

| Input | Result |
|---|---|
| 1<br>3 10 17 57<br>6<br>2 7 10 15 57 246 | 10 57 |

**Answer:** (penalty regime: 0 %)

```
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int a;
    scanf("%d",&a);
    int A[a];
    for(int i=0;i<a;i++){
        scanf("%d",&A[i]);
    }
    int b;
    scanf("%d",&b);
    int B[b];
    for(int i=0;i<b;i++){
        for(int i=0;i<b;i++){
            scanf("%d",&B[i]);
        }
    }
    for(int i=0;i<a;i++){
        for(int j=0;j<b;j++){
            if(A[i]==B[j]){
                printf("%d ",A[i]);
```

```
            printf("%d ",A[i]);
        }
    }
}
return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 1<br>3 10 17 57<br>6<br>2 7 10 15 57 246 | 10 57 | 10 57 | ✔ |
| ✔ | 1<br>6 1 2 3 4 5 6<br>2<br>1 6 | 1 6 | 1 6 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Benno Jefrina J 251001028-2025-IT

B2

| | |
|---|---|
| **Started on** | Thursday, 17 September 2026, 11:11 AM |
| **State** | Finished |
| **Completed on** | Thursday, 17 September 2026, 11:12 AM |
| **Time taken** | 48 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

**For example:**

| Input | Result |
|---|---|
| 3<br>1 3 5<br>4 | 1 |

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

    int a[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &a[i]);
    }

    int k;
    if (scanf("%d", &k) != 1) return 0;

    int i = 0;
    int j = 1;
    int found = 0;

    while (j < n) {
        int diff = a[j] - a[i];

        if (diff == k && i != j) {
            found = 1;
            break;
        } else if (diff < k) {
            j++;
        } else {
            i++;
        }

        if (i == j) {
            j++;
        }
    }

    printf("%d", found);
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 3<br>1 3 5<br>4 | 1 | 1 | ✔ |
| ✔ | 10<br>1 4 6 8 12 14 15 20 21 25<br>1 | 1 | 1 | ✔ |
| ✔ | 10<br>1 2 3 5 11 14 16 24 28 29<br>0 | 0 | 0 | ✔ |
| ✔ | 10<br>0 2 3 7 13 14 15 20 24 25<br>10 | 1 | 1 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Benno Jefrina J 251001028-2025-IT

B2

| | |
|---|---|
| **Started on** | Thursday, 17 September 2026, 11:12 AM |
| **State** | Finished |
| **Completed on** | Thursday, 17 September 2026, 11:12 AM |
| **Time taken** | 15 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

**For example:**

| Input | Result |
|---|---|
| 3<br>1 3 5<br>4 | 1 |

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

    int a[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &a[i]);
    }

    int k;
    if (scanf("%d", &k) != 1) return 0;

    int i = 0;
    int j = 1;
    int found = 0;

    while (j < n) {
        int diff = a[j] - a[i];

        if (diff == k && i != j) {
            found = 1;
            break;
        } else if (diff < k) {
            j++;
        } else {
            i++;
        }

        if (i == j) {
            j++;
        }
    }

    printf("%d", found);
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 3<br>1 3 5<br>4 | 1 | 1 | ✔ |
| ✔ | 10<br>1 4 6 8 12 14 15 20 21 25<br>1 | 1 | 1 | ✔ |
| ✔ | 10<br>1 2 3 5 11 14 16 24 28 29<br>0 | 0 | 0 | ✔ |
| ✔ | 10<br>0 2 3 7 13 14 15 20 24 25<br>10 | 1 | 1 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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